SCHOOL OF COMPUTER SCIENCE & ELECTRICAL ENGINEERING, UNIVERSITY OF SURREY

NATURAL LANGUAGE PROCESSING (COMM061), SEMESTER 2, 2025-26

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GROUP COURSEWORK

Submission deadline: Week 11; 6th May, Wednesday, 4 PM.

General Guidelines

This coursework constitutes 50% of the overall assessment for COMM061. The assessment will be based on the criteria specified in relevant sections of this document, where the appropriate mark carried by an individual part of a question is indicated in brackets ( ) and <u>are underlined</u>.

The coursework will include the submission and evaluation of the following elements:

- Coursework report in PDF format submitted on *SurreyLearn*;
- Source code to be submitted to *SurreyLearn* in a ZIP;

<u>The coursework report should be named as follows: report_<GroupName>.pdf.</u> The coursework report should contain a page with a declaration of originality as shown in My Surrey -> Exams and Assessments -> Assessments. The e-signature can be a scanned picture of your printed signature, a digital signature drawn on computer, or a signature added using Adobe Acrobat's signing feature.

Extensions, Late Submissions and Academic Integrity:

- Coursework will be routinely checked for academic misconduct:
- Please refer to your Student Handbook and the advice given on *SurreyLearn* on plagiarism and collusion and make sure that you understand the regulations.
- If you are in any doubt, please seek advice from your Module Leader or Personal Tutor.
- Both the report and the source code should be submitted electronically by the **deadline**. Pay attention to late submission policies of the University: late submissions will lead to 10% penalty per day up to 48-hours (2 calendar days) after the deadline, for submissions between 48-hours (2 calendar days) and 120-hours (5 calendar days), the mark given for that unit of assessment will be the Pass mark (40%), and any submission from 4.01pm on the fifth day after the deadline (120 hours), or non-submissions will receive 100% penalty (i.e., it will not be assessed, and a grade of 0% will be automatically awarded).

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Topic Overview

Sequence classification is the task of assigning a predefined class label to input text. It serves important NLP applications like customer feedback analysis, identification of socio-linguistic constructs (offense, hate, sarcasm), content moderation and so on. This coursework requires you to apply it to global varieties of English. Large Language Models (LLMs) treat English as a 'monolith' and often fail to capture the nuances of non-standard varieties. This can lead to significant bias in downstream tasks requiring cultural context like detecting sarcasm, translating dialectal idioms, interpreting regional sentiment. In this coursework, we attempt to address the challenge of dialectal robustness. You will build models for Sentiment Analysis and Sarcasm Detection for three varieties of English and work together as a team to understand whether models trained on one variety of English struggle to generalize to another. You are strongly encouraged to explore this "variety gap" and manually analyse erroneous predictions made by your models.

You will use the BESSTIE dataset, a benchmark designed to evaluate these biases across three varieties: British (en-UK), Australian (en-AU), and Indian (en-IN). This data is sourced from Google Places reviews and Reddit comments and includes two "Inner-Circle" varieties (UK/Australian) and one "Outer-Circle" variety (Indian). You may encounter linguistic challenges, from code-mixed Hindi-English in en-IN to locale-specific colloquialisms in en-AU and en-UK. In this dataset, tasks are modelled as simple binary classification problems:

- Sentiment: Labeled as 1 for Positive and 0 for Negative.
- Sarcasm: Labeled as 1 for Sarcastic and 0 for Not Sarcastic.

The data contains linguistic features such as local slang (*e.g.,* "arvo" , "benstokes"), dialectal grammar, and code-mixed vocab. Table below shows example segments from the data.

Variety	Text	Sentiment	Sarcasm
en-AU	"Absolute legend, parked his ute right across my driveway. Good onya, mate."	0	1
en-IN	"Coz we all have free internet."	0	1
en-UK	"Traditional friendly pub. Excellent beer"	1	0

Your task is to build and compare different sequence classification models using this dataset. You will explore various NLP approaches, including classical machine learning (*e.g.,* SVMs, Logistic Regression), fine-tuned BERT-like autoencoder models (*e.g.,* BERT, RoBERTa), and potentially LLMs using zero-/few-shot prompting. You will evaluate the performance of each approach using metrics such as Macro-F1 score.

Dataset Link: https://huggingface.co/datasets/surrey-nlp/BESSTIE-CW-26

With this coursework, the aim is not to just perform the sequence classification task but explore different NLP approaches; while trying to develop an understanding of the linguistic/real-world challenges involved in building culturally sensitive NLP/AI systems.

NOTE: You are provided a deduplicated version of the BESSTIE dataset which contains specific splits (Train, Validation, Test) for each variety. For experimentation, ensure you generally respect these splits to allow for fair comparison. You must not modify the test split. Please note that sarcasm instances face a class imbalance given the real-world nature of the data. You are encouraged to treat data for this imbalance.

Structure

The module is assessed 50% on this group coursework. The coursework is divided into two parts requiring a group to make one single submission:

1. A group formation and plan of work (not assessed).
2. An experimentation and deployment part, based on the group plan. The outcome of this part is the **final report (max. 25 pages content excluding references)** and **the codebase**.

You will be grouped on *SurreyLearn* and must be aware of the team members from the shared sheet. It is highly recommended you start discussions as early as possible and come up with *a concrete plan*.

The methodology is much more important than the accuracy of the model. So, make sure you follow the taught methods appropriately for your experiments rather than spending much time improving accuracy.

Work continuously as a group, ensuring you as an individual contribute to the team. The team should be there to support any of the members facing issues, and in exceptional cases, changes to the originally submitted plan will be allowed.

Although you are allowed (and I encourage you) to search the Web for finding examples, code snippets, and to use teaching material such as lab notebook code, data, definitions, diagrams, lecture slides, directly or indirectly in your submissions - all such material must be referenced concisely.

No additional infrastructure should be requested for coursework experiments. You are to use GPUs provided in the Heron lab, Babbage cluster, or free tier cloud GPUs.

None of the material in your submission, *viz.*, code, written text, comments for code, report text and so on, should be generated via language model(s) (*e.g.,* GPT/Gemini writing your report/code). Exploring LLMs as part of the coursework experiments is obviously permitted, but the submissions must be your own work. Any such instance will be considered a case of academic misconduct and is liable to be reported to the University immediately.

Group Declaration

Weight: 0% of the marks
Submission date: Week 6 (23.59, Friday, 13th March, *SurreyLearn* Upload, only one per group)

This first part is important as it will set the group and work structure for the next two parts. Discuss as a group and decide on the following:

1. Discuss and gain a collective understanding of the sequence classification task. Ensure you understand the binary labelling schema for Sentiment (0/1) and Sarcasm (0/1). You must also ensure you understand the *three language varieties* present in the BESSTIE dataset: en-AU, en-IN, and en-UK.
2. Discuss the distribution of instances in each split of the BESSTIE dataset (Train, Validation, and Test) and across the two domains (Google Places *vs.* Reddit).
3. Discuss your strategy for handling the "Variety Gap".
    - THINK: Will you treat all data as one large pool, or will you strictly separate training data by variety (*e.g.,* training on UK only) to test robustness?
4. Plan experiments, while keeping in mind your resources (GPU hours) / time constraints and concretize the scope and priorities.
5. Decide on a common development environment (code repository, Python libraries needed like transformers or scikit-learn, *etc.*).
6. Divide individual tasks and explain how you are planning to work as a team. Each group member could essentially focus on a different aspect of the pipeline. This can be experimenting with different:
    - Data Analysis and Visualisation.
    - Implementing classical models as baselines.
    - Cross-Variety Evaluation, i.e., training of one variety and testing on another.
      Fine-tuning with different pre-trained models.
    - Manual error analysis of erroneous predictions and explanation

Deliverables for this declaration Each group will need to submit a PDF document (max. two pages, min size 10 font) including member details (group member names, URNs), a (very brief) summary of dataset statistics observed, the plan for the experiments, the development environment, and the individual tasks per group member.

No front cover or appendix is needed.

This needs to be submitted by ONE of the group members on *SurreyLearn*, before the deadline.

The one-page document MUST be named: PG<number>.pdf, *e.g.*, for PG01 from the shared sheet, PG01.pdf. Please ensure that the group member who submits this PDF follows the naming convention, and only 1 member uploads it.

Group Submission

Weight: 50% of the module marks
Submission date: Week 11; 6th May, Wednesday, 4 PM.

Experimentation and Evaluation (80 marks)

The group should research and experiment on different ways to train a predictive model for **Sentiment** and **Sarcasm classification** across varieties of English. The choice of different individual workloads (tasks) should be discussed and decided upon as a group, as per your group declaration report.

All experiments and documentation must be done in Python notebook(s). It is recommended you submit a single notebook that clearly delineates each experiment with headers. If you choose to submit multiple notebooks, ensure there is a *main.ipynb* that executes your code. Please ensure your report PDF is structured and page-limited as per the questions below.

1. Analyse and visualise the dataset (15 marks; max. 4 pages). Produce charts and document your observations.

1.1. Visualise different distributions and comment on the class imbalance between Sentiment (Positive/Negative) and Sarcasm (Sarcastic/Not Sarcastic) across all three varieties (en-AU, en-IN, en-UK). [5 marks]

[THINK: As a computational linguist, what aspects of this dataset might be interesting to observe apart from simple label distributions- Parts-of-speech, dependency parse of sarcastic phrases, or tokenization of some slang?]

1.2. Vocabulary Analysis. You are encouraged to use papers in the references section below (and other relevant literature) to understand how you can analyse a dataset for sequence classification. [10 marks]

- Calculate the vocabulary overlap between any one of the inner-circle (en-UK/en-AU) and outer-circle varieties (en-IN). [THINK: Is the Indian English subset significantly different from the British/Australian subset?]
    - HINT: Jaccard Similarity or Cosine Similarity of TF-IDF vectors
- You must include a short paragraph in your answer which defines "linguistic distance" and whether this may affect model performance or is this "variety distance" just superficial and driven by topic-specific nous rather than variations in grammar.

2. Experimentation with three different setups (40 marks, 10 marks per setup; max. 6 pages). Your PDF report should have a subheading for each experiment, numbered 2.1, 2.2, and 2.3. You MUST HAVE two or more training/tuning runs per setup to ensure your results are not due to random chance.

- 2.1. Baseline/PTLM Gap [10 Marks]
    - Compare the performance of a classical baseline against a fine-tuned Transformer for the **Sentiment/Sarcasm** Task.
        - HINT: TF-IDF/GloVe + Logistic Regression/SVM *vs.* RoBERTa-base
- 2.2. Cross-variety Evaluation [15 Marks]

    - Perform full Cross-Variety Evaluation Matrix using any one pre-trained encoder language model for any one task (**Sentiment/Sarcasm**). Train/Fine-tune on each variety individually and test on all others. Analyse for transferability. THINK: Does an encoder model fine-tuned on Australian English transfer better to British English than to Indian English?
  - You may also use existing implementations like LIME/SHAP for interpretable evaluation of the errors made by models.
- 2.3. LLM Variations [15 Marks]
    - Use LoRA (Low-Rank Adaptation) to train lightweight adapters for each variety using LLMs for the **Sarcasm** task. Compare the performances on each test set *viz.* "UK Adapter" *vs.* an "Indian Adapter" with the same frozen base LLM. You are allowed to modify the training set size to reduce time taken for each variation.
    - HINT: You MUST use an open-weight LLM in parameter range of 1B – 3B.

3. **Evaluation (15 marks; max. 5 pages).** Analyse testing for each of the three experiment setup variations conducted above.

- **Metrics.** Report Macro-F1 score as the primary metric but precision and recall must be included in report. Per-class results must be shown and discussed.
- **Visuals.** Provide Confusion Matrices for your best performing models to show exactly where they fail.
- Do not just report one number for each experiment. You must report performance specifically for the "sarcasm" *vs.* the "non-sarcasm" OR "positive" *vs.* "negative" classes. Do note that a model with 90% accuracy that never predicts "sarcasm" is a failure; you must show your best model is not just predicting the majority class.

4. **Sarcasm Explanation & Error Analysis (10 marks; max. 4 pages).** Do not just show wrong predictions, do the below:

- Extract any 10 instances of erroneous predictions made by the best LLM/adapter.
- Choose any 4 examples for which you can explain why they are/not sarcastic; and add explanations to prepare a single Few-Shot Prompt with these 4 examples paired with their correct ground-truth labels and your explanations.
- Now test the remaining 6 examples with THIS few-shot prompt, including example labels and your explanations for the 4 above. Also, show your analysis with new results.
- Comment on whether any erroneous predictions have improved given the few-shot prompt or not.

Deployment (20 marks)

5.1. **Deployment Endpoint (15 marks; max. 5 pages).**

- Build a web service to host your best-performing model. You can use Gradio, Streamlit, or Flask to do this. UI must allow the user to input text and explicitly select the English variety.
  - Based on the user's selection, the backend should switch to the appropriate model (or LoRA adapter) to give the best prediction for that context. If "British" is selected, the system loads the specific LoRA/model weights for British English.
  - HINT: You may host your LoRA/model on Huggingface and load it directly into your application using the *transformers* library [via *from_pretrained()*], avoiding large file uploads in your submission ZIP.
  - You can show screenshots.

- Briefly explain why you chose this deployment/serving method.
  - HINT: If you used LoRA adapter, you may discuss the advantage of swapping small adapter weights versus reloading an entire model for every user request.

5.2. **Efficiency (5 marks; max. 1 page).**

- Compute the average inference time (in milliseconds) for two of your models on a standard input, and some subsequently large inputs/batches. Discuss the trade-off between Model Size (*e.g.,* RoBERTa-base vs SVM) and Response Time. Would you sacrifice performance for speed, or vice-versa?

Deliverables

You will need to submit a Jupyter notebook documented appropriately, and a PDF report:

1. ".ipynb" notebook file (plus any helper files you might use) [mandatory File 1, or inside a ZIP with other files where the main notebook is *main.ipynb*]
2. PDF report (max. 25 pages content excluding references), with each answer described following details from above. It should contain either the results in tables or screenshots of results from notebook. It must have a sample of your error analysis of mispredictions made by the best model, and any other relevant sections based on [mandatory File 2, outside ZIP]

Deliverable Checklist

- The notebook should contain visuals (where appropriate) to support tasks such as: label data distribution, histogram comparisons, text samples, classification accuracy curve, confusion matrix, *etc.* Additional notebooks or Python files can also be included, but make sure you zip the files together before submitting. If there are library dependencies, please also include a requirements file.
- Only zip the code related file together, DO NOT put the pdf report in the zip file, so submit this separately, so that it can be checked for plagiarism.
- DO NOT print the entire dataset or any list/dictionary etc on the notebook. Instead, just print the top few records.
- DO NOT submit the dataset(s), just add reference(s)
- DO NOT submit the trained model(s), MUST not include the /runs or /checkpoints folders.

References

Srirag, Dipankar, Aditya Joshi, Jordan Painter, and Diptesh Kanojia. "BESSTIE: A benchmark for sentiment and sarcasm classification for varieties of english." In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 8413-8429. 2025. [https://aclanthology.org/2025.findings-acl.441/]

Gavin Abercrombie and Dirk Hovy. 2016. Putting Sarcasm Detection into Context: The Effects of Class Imbalance and Manual Labelling on Supervised Machine Classification of Twitter Conversations. In *Proceedings of the ACL 2016 Student Research Workshop*, pages 107–113, Berlin, Germany. Association for Computational Linguistics. [https://aclanthology.org/P16-3016/]

Singh, Ishmanbir, Dipankar Srirag, and Aditya Joshi. "Nek Minit: Harnessing Pragmatic Metacognitive Prompting for Explainable Sarcasm Detection of Australian and Indian English." *arXiv preprint arXiv:2505.15095* (2025). [https://arxiv.org/abs/2505.15095]

Barbara Plank. 2022. The "Problem" of Human Label Variation: On Ground Truth in Data, Modeling and Evaluation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 10671–10682, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics. [https://aclanthology.org/2022.emnlp-main.731/]

Stephen Skalicky and Scott Crossley. 2018. Linguistic Features of Sarcasm and Metaphor Production Quality. In *Proceedings of the Workshop on Figurative Language Processing*, pages 7–16, New Orleans, Louisiana. Association for Computational Linguistics. [https://aclanthology.org/W18-0902/]

Frenda, Simona, Gavin Abercrombie, Valerio Basile, Alessandro Pedrani, Raffaella Panizzon, Alessandra Teresa Cignarella, Cristina Marco, and Davide Bernardi. "Perspectivist approaches to natural language processing: a survey." Language Resources and Evaluation 59, no. 2 (2025): 1719-1746.

LIME: 1) Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "Model-agnostic interpretability of machine learning." *arXiv preprint arXiv:1606.05386* (2016).
2) Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. ""Why should I trust you?" Explaining the predictions of any classifier." In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pp. 1135-1144. 2016.
[HELP: https://christophm.github.io/interpretable-ml-book/lime.html]
[TOOL: https://github.com/marcotcr/lime]

SHAP: Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." *Advances in neural information processing systems* 30 (2017).
[HELP: https://christophm.github.io/interpretable-ml-book/shap.html]
[TOOL: https://github.com/shap/shap]

Group Engagement & Discussion

If for any reason one of the team members is unable to contribute, as a group you must flag them to an instructor (DK/SD). In case you are emailing an unresponsive member of the group, you MUST CC the module lead for information. Members having not contributed to the submission will not be provided ANY MARKS. Therefore, it is suggested to ensure regular follow-ups within the group for the work you divide amongst each other. Please maintain contact and coordinate with team members during the semester. It is recommended you use discussion forum on *SurreyLearn* to discuss any relevant points with your team, ask questions about the coursework and get clarity in advance regarding any doubts you may have.

Rubric

Experimentation (80 marks)

CW-1	Marks	Criteria.
Q1	13-15	Extensive analysis of the dataset(s) containing clear visuals, observations, and explanations, novel findings (*e.g.,* slangs OR mistakes in dataset), good commentary on domain-specificity and linguistic distance.
	8-12	Good analysis of the data, containing visuals and explanations, enough to provide confidence on what is expected, good but limited commentary on variety/domain.
	4-7	Some analysis has been done, including visuals, but additions work was needed to ensure the dataset is fit for purpose, minor misinterpretations in analysis.
	1-3	Only images present, no analysis but actual distribution plots present.
	0	No analysis present, incorrect visuals and graphs, majorly misinterpreted analysis.

Q2.1	9-10	Excellent comparison between classical and transformer models. Insightful justification of why pre-training helps (or doesn't) with sarcasm.
	6-8	Good choice of a setup, and good implementation of both baselines. Results are clear, but the analysis of why the gap exists is superficial.
	3-5	The setup is valid, but the variation chosen is still not representative to fully understand the correct setting or Unclear documentation across code base.
	0-2	The experimental setup is very unclear or is redundant or is invalid /does not contribute to any evaluation.
Q2.2	12-15	Flawless/Good execution of the full evaluation matrix. Deep analysis of transferability. Clear handling of train/test splits.
	7-11	Matrix is complete but analysis is descriptive ("A is better than B") rather than analytical. Issues with data or split management.
	1-6	Matrix is incomplete or the experimental design for cross-variety testing is invalid.
	0	No answer provided or no submission
Q2.3	12-15	Technical novelty. Successful implementation of LoRA adapters or advanced data augmentation. Clear proof of efficiency or robustness gains.
	7-11	Good implementation of the advanced technique but lacks critical evaluation of its benefits.
	1-6	Only one or two LoRA implemented, not all three, but there is some comparative analysis present.
	0	No answer provided or no submission

Q3 (5 x 3)	4-5	Comprehensive testing and good explanation of the test results with supporting visuals, which indicates if a model/technique is adequate.
	2-3	Some testing was done, but it does not really offer a certain picture of the model performance. The visuals used were limited.
	0-1	Not much testing was conducted, and the explanation was not appropriate. There were no correct visuals such as graphs to show results, or incorrect evaluation.

Q4	8-10	Excellent error analysis and prompt engineering. The 4-shot prompt includes clear, linguistically sound explanations. Excellent critical comparison of the “before and after” results for the remaining 6 examples.
	4-7	Good attempt at few-shot prompting, but the explanations in the prompt are generic. The “before and after” analysis is present but lacks depth.
	1-3	Very basic error analysis. The few-shot prompt was either not implemented correctly, o the student failed to evaluate if the new prompt improved the predictions.
	0	No error analysis or few-shot prompting attempted.

Deployment (20 marks)

	Marks	Criteria
Q5.1	13-15	Great architecture choice and implementation of the service. The model/mechanism can deploy easily and serve their purpose efficiently. Example: Many sample predictions shown within a flexible framework which can deploy different models.
	8-12	Good architecture and efficient implementation of the core mechanism but needed some additional functionality to allow for easier deployment, needs better front-end.
	4-7	Working implementation of the service, although not the most efficient. Some parts contained major errors in deployment, or execution of service.
	1-3	There was not much implementation or there were many errors, and the model/mechanism would not deploy.
	0	No submission for deployment part.

Q5.2	4-5	Clear measurement of inference time. Insightful discussion on the trade-off between model size and latency.
	2-3	Measurements provided but lacks critical discussion of the trade-offs.
	1	Implementation is in code, but no efficiency analysis provided in report.
	0	No answer or submission